

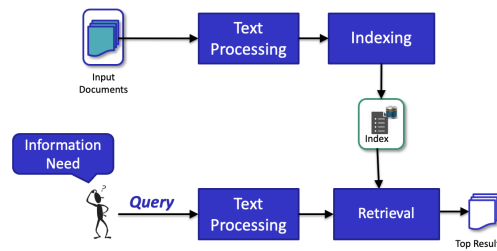
Cribsheet

General

Log Table

x	$\log_{10}(x)$	x	$\log_{10}(x)$	x	$\log_{10}(x)$	x	$\log_{10}(x)$	x	$\log_{10}(x)$	x	$\log_{10}(x)$	x	$\log_{10}(x)$
0.10	-1.0	0.60	-0.2	1.25	0.1	2.50	0.4	4.5	0.7	7.0	0.8	9.5	1.0
0.20	-0.7	0.70	-0.2	1.50	0.2	2.75	0.4	5.0	0.7	7.5	0.9	10.0	1.0
0.30	-0.5	0.80	-0.1	1.75	0.2	3.0	0.5	5.5	0.7	8.0	0.9	15.0	1.2
0.40	-0.4	0.90	0.0	2.00	0.3	3.5	0.5	6.0	0.8	8.5	0.9	20.0	1.3
0.50	-0.3	1.00	0.0	2.25	0.4	4.0	0.6	6.5	0.8	9.0	1.0	25.0	1.4

Architecture



Weighting

Heap's Law: $V = Kn^\beta$, typically $K \approx 10 - 100$, $\beta \approx 0.5$

$$idf(t_k) = \log \left(\frac{N}{D_k} \right)$$

Vector Space Model

$$\text{Cosine}(D_i, Q) = \frac{\sum_{j=1}^t d_{ij} \cdot q_j}{\sqrt{\left(\sum_{j=1}^t d_{ij}^2 \right) \left(\sum_{j=1}^t q_j^2 \right)}}$$

Bayes rules in Language Modelling

$$p(D | Q) = \frac{p(D)p(Q | D)}{p(Q)} \stackrel{\text{rank}}{=} p(D)p(Q | D)$$

Jelinek Mercer Smoothing in Language Modelling

$$P(q|d) = \prod_{w \in q} (1 - \lambda) \cdot P(w|d) + \lambda \cdot p(w|C) \quad (1)$$

$$= \prod_{w \in q} (1 - \lambda) \cdot \frac{tf_{w,d}}{|d|} + \lambda \cdot \frac{tf_{w,C}}{|C|} \quad (2)$$

Per-fields normalisation

$$\text{score}(d, Q) = \sum_{t \in Q} w \left(t f n = \sum_f w_f \cdot \text{Norm}(t f_f) \right)$$

Evaluation

$$\text{Recall} = \frac{|\text{relevant} \cap \text{retrieved}|}{|\text{retrieved}|} \quad \text{Precision} = \frac{|\text{relevant} \cap \text{retrieved}|}{|\text{retrieved}|} \quad F_1 = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$\text{DCG}_p = \sum_{i=1}^p \frac{\text{rel}_i}{\log_2(i+1)} = \text{rel}_1 + \sum_{i=2}^p \frac{\text{rel}_i}{\log_2(i+1)} \quad \text{nDCG} = \frac{\text{DCG}}{\text{IDCG}}$$

Relevance Feedback

Rocchio:

$$\mathbf{q}_{\text{new}} = \alpha \mathbf{q}_0 + \frac{\beta}{|D_r|} \sum_{\mathbf{d}_j \in D_r} \mathbf{d}_j - \frac{\gamma}{|D_n|} \sum_{\mathbf{d}_j \in D_n} \mathbf{d}_j$$

PyTerrier operators

Op.	Name
>>	Then
%	Rank Cutoff
	Set Union
&	Set Intersection
**	Feature-Union

Neural IR

$$\text{MaxSim} = \sum_{q_i \in Q} \max_{d_i \in D} \text{sim}(q_i, d_i)$$

